

Crystal forest

If you dip a piece of pure metal into a solution in which another metal is dissolved, something quite magical may happen.

These delicate crystals have formed on a piece of zinc placed in a solution of lead nitrate. The magic is in fact a chemical reaction known as metal displacement, seen here in a photograph taken through a microscope. The more reactive metal (zinc) displaces the less reactive metal (lead) from its nitrate compound, so instead of lead nitrate and zinc, we end up with pure lead and a solution of zinc and nitrate ions. The lead atoms join together in regular patterns, forming crystals of pure lead.

